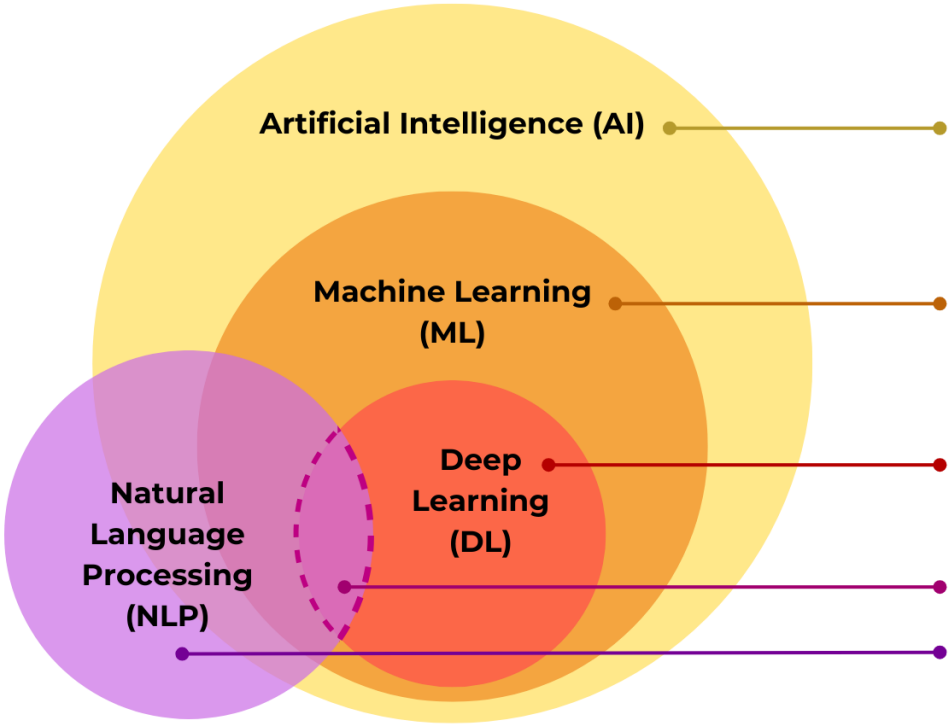
Artificial Intelligence vs Human Intelligence
Key concepts of AI



Machines that perform tasks requiring “human-like” intelligence

Methods for “teaching” machines to perform a task using examples (data)

A particular ML method involves tuning parameters in a “neural network” structure

Large Language Models

Methods to analyze, understand, and generate human language

What is Human Intelligence?

Human Intelligence is the **natural ability of humans to think, learn, reason, and make decisions based on knowledge and experience.**
It helps people understand situations, solve problems, and adapt to new environments.

Key Abilities of Human Intelligence.

Creativity

- **Ability to generate new ideas and innovative solutions.**

Example: An entrepreneur developing a new business strategy.

Emotional Understanding

- **Ability to recognize and manage emotions in oneself and others.**

Example: A manager motivating employees during difficult situations.

Experience-Based Learning

- **Humans learn from past experiences and improve decision-making.**

Example: A salesperson improving negotiation skills after dealing with many clients.

Adaptability

- **Ability to adjust to new situations and challenges.**

Example: Businesses shifting from offline stores to online platforms.

What is Artificial Intelligence?

Artificial Intelligence (AI) refers to the ability of **machines or computer systems to perform tasks that normally require human intelligence**, such as learning, reasoning, and decision-making.

Examples of Artificial Intelligence

Voice Assistants

- **AI systems that respond to voice commands.**

Example: Siri, Alexa.

Recommendation Systems

- **Suggest products, movies, or services based on user preferences.**

Example: Netflix or Amazon recommendations.

Chatbots

- **AI programs that interact with customers and answer queries automatically.**

Example: Customer support chatbots on websites.

Fraud Detection

- **AI systems used by banks to identify suspicious transactions.**

Example: Detecting unusual credit card activity.

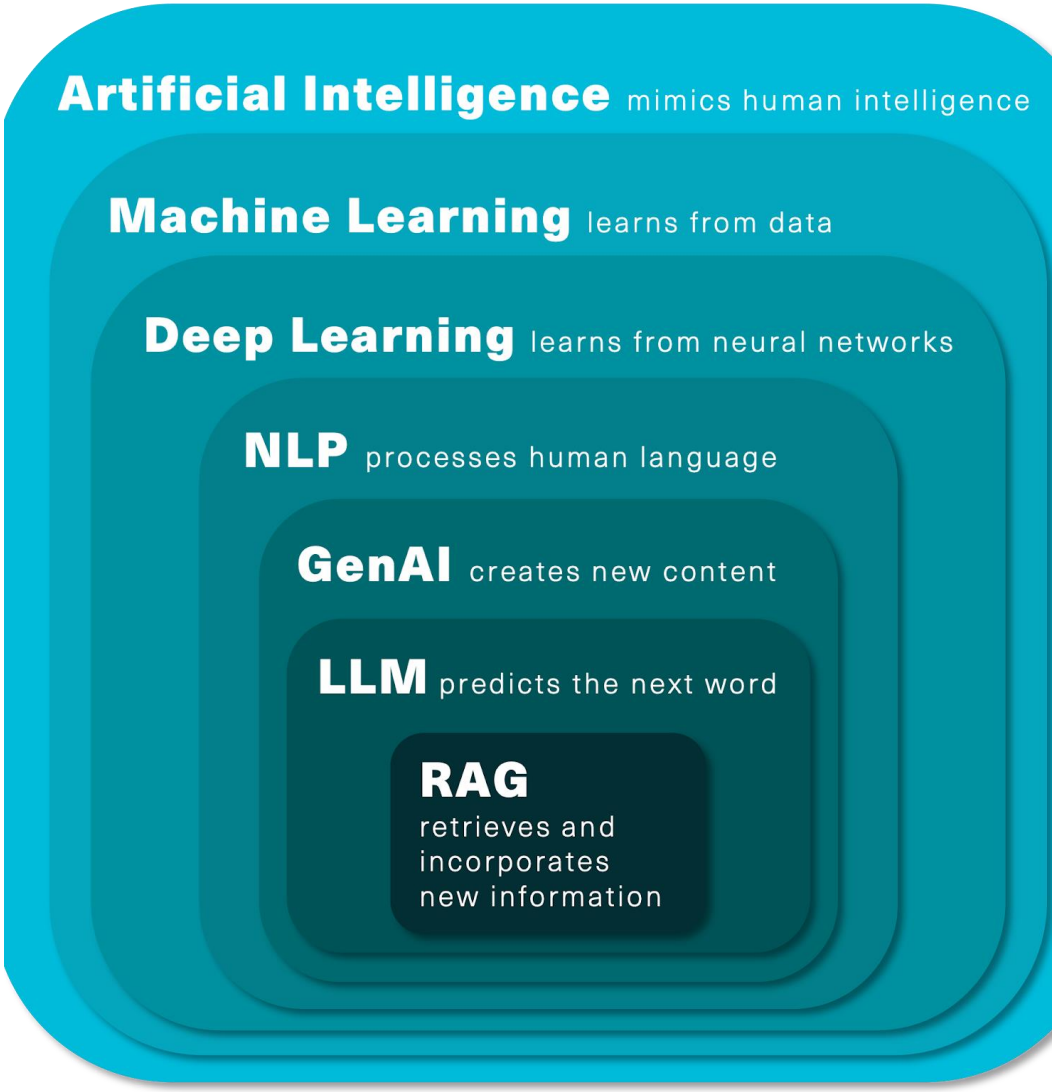
AI vs Human Intelligence

Feature	Human Intelligence	Artificial Intelligence
Nature	Natural intelligence developed in humans	Machine-based intelligence created by humans
Learning	Learns from experience, education, and observation	Learns from data using algorithms and models
Creativity	Highly creative and innovative	Limited creativity (depends on programming)
Emotions	Has emotions and empathy	No emotions or feelings
Decision Making	Based on experience, intuition, and judgment	Based on data analysis and programmed rules
Speed	Slower in processing large data	Very fast in processing large datasets
Adaptability	Highly adaptable to new situations	Limited adaptability
Example	Manager making strategic decisions	AI system recommending products online

Key AI Concepts(Technologies)

Important AI technologies include:

- Machine Learning
- Deep Learning
- Natural Language Processing.



Machine Learning

Machine Learning (ML)

Machine Learning is a branch of Artificial Intelligence that enables **computers to learn from data and improve their performance without being explicitly programmed.**

Instead of following fixed instructions, the system **analyzes patterns in data and makes predictions or decisions.**

Business Example

Online Shopping Platforms

Online stores use Machine Learning to **recommend products to customers** based on:

- Previous purchases
- Browsing history
- Customer preferences

Example: If a customer buys a **mobile phone**, the system may recommend **phone cases, earphones, or screen protectors.**

Types of Machine learning

Machine Learning is generally classified into **three main types based on how the system learns from data.**

1. Supervised Learning

The model is trained using **labeled data** (data with known answers).
The system learns the relationship between input and output.

Example:
Email spam detection where emails are labeled as spam or not spam.

2. Unsupervised Learning

The model learns from **unlabeled data** and discovers hidden patterns or relationships.

Example:
Customer segmentation in marketing where businesses group customers based on buying behavior.

3. Reinforcement Learning

The system learns by **interacting with the environment** and receiving **rewards or penalties**.

Example:
AI systems used in game playing or robotics.

Deep Learning

Deep Learning is a **subset of Machine Learning** that uses **Artificial Neural Networks with multiple layers** to learn patterns from large amounts of data.

It is inspired by the **structure of the human brain**, where neurons process and transmit information. In deep learning, these layers of artificial neurons help the system automatically learn complex patterns and make accurate predictions.

Key Characteristics

- Uses **multi-layer neural networks**
  - **Learns automatically from large datasets**
  - **Very effective for complex data such as images, audio, and text**

Examples of Deep Learning

Image Recognition

- **Identifying objects or faces in images.**

Example: Face unlock feature in smartphones.

Speech Recognition

- **Converting spoken language into text.**

Example: Voice assistants like Siri or Alexa.

Autonomous Vehicles

- **Self-driving cars use deep learning to recognize roads, pedestrians, and traffic signals.**

Natural Language Processing

Natural Language Processing (NLP)

Natural Language Processing (NLP) is a **field of Artificial Intelligence that focuses on enabling computers to understand and process human language.**

NLP systems often use **Machine Learning and Deep Learning techniques** to analyze text and speech.

Examples.

- Chatbots used in customer service
  - Voice assistants like Siri or Alexa
  - Language translation systems
  - Sentiment analysis of customer reviews

Business Example:

Companies analyze customer feedback and social media comments using NLP to understand customer opinions.

AI vs Machine Learning vs Deep-Learning in Business

Artificial Intelligence (AI)

Broad field that enables machines to perform tasks that normally require human intelligence.

Chatbots used by companies to handle customer queries.

Machine Learning (ML)

A subset of AI where systems learn patterns from data and improve automatically.

Online stores recommending products based on customer purchase history.

Deep Learning (DL)

A subset of Machine Learning that uses neural networks to analyze complex data such as images and speech.

Banks using facial recognition for secure authentication.

AI technologies in real-world business applications

AI Technology	Description	Real-World Business Applications	Example
Machine Learning (ML)	Technology that allows computers to learn patterns from data and make predictions without explicit programming.	Product recommendation, fraud detection, customer segmentation, sales prediction.	E-commerce websites recommending products based on customer purchase history.
Deep Learning (DL)	Advanced form of machine learning that uses multi-layer neural networks to analyze complex data such as images, audio, and video.	Facial recognition, speech recognition, medical image analysis, security systems.	Banks using facial recognition for secure authentication.
Natural Language Processing (NLP)	AI technology that enables computers to understand and process human language in text or speech form.	Chatbots, sentiment analysis, language translation, voice assistants.	Companies analyzing customer reviews on social media to understand customer opinions.